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Diagnostic Accuracy of Plasma 1,5-Anhydroglucitol for Detecting Prediabetes in Obese Young Adults

ABSTRACT

Prediabetes is a transitional metabolic condition characterized by ⁹ blood glucose levels that are elevated above normal but do not yet meet the diagnostic criteria for diabetes mellitus. Identifying this condition at an early stage is particularly important among obese young adults, who are at a heightened risk of developing metabolic complications. Common diagnostic approaches, including fasting blood ⁴² glucose, oral glucose tolerance test, and glycated hemoglobin (HbA1c), present several limitations related to physiological variability and reduced reliability in specific clinical situations. Consequently, there is a growing need for alternative biomarkers capable of detecting early disturbances in glycemic control.

This study employed a diagnostic accuracy design conducted over six months at Dr. M. Djamil General Hospital, Padang, Indonesia. A total of 143 obese individuals aged 18–40 years were recruited through incidental sampling. Blood samples were collected to determine HbA1c and plasma ³⁹ 1,5-Anhydroglucitol (1,5-AG) levels, with HbA1c serving as the reference standard for classification. The diagnostic performance of 1,5-AG was assessed using sensitivity, specificity, and predictive values.

Among the participants, 20 individuals (14.0%) were categorized as having prediabetes. Plasma 1,5-Anhydroglucitol demonstrated a sensitivity of 90.0%, specificity of 74.8%, ³² positive predictive value of 36.7%, and negative predictive value of 97.9%.

Overall, plasma 1,5-AG showed strong sensitivity and an excellent ability to exclude prediabetes, as reflected by its high negative predictive value. ¹⁹ These findings indicate that 1,5-AG may serve as a useful complementary biomarker, particularly for screening purposes, although confirmation using established glycemic indicators remains necessary.

Keywords: Prediabetic state; obesity; young adult; 1,5-Anhydroglucitol

Prediabetes represents an intermediate state of glucose dysregulation in which circulating glucose levels exceed normal values but remain below the threshold required for a diagnosis of diabetes mellitus. The concept was formally introduced by the Department of Health and Human Services (DHHS) and the American Diabetes Association (ADA) in 2002, and continues to be refined in the most recent ADA Standards of Medical Care in Diabetes (2025). Currently, prediabetes poses a substantial global health challenge, affecting a large proportion of the population. It was estimated that approximately 314 million individuals were living with prediabetes in 2019, with projections indicating an increase to 418 million by 2025. Without timely intervention, around 4–9% of affected individuals may progress to diabetes each year.¹² In the United States, data from the Centers for Disease Control and Prevention (CDC) suggest that nearly one in three adults may have prediabetes, many of whom are unaware of their condition. A similar upward trend has been observed in Indonesia, where the prevalence reached 30.8% in 2018. Early identification of prediabetes is therefore crucial, especially in young adults, as early metabolic disturbances may contribute to long-term complications such as type 2 diabetes and cardiovascular disease.^{3–6}

Various diagnostic tools are currently used to detect prediabetes, including fasting blood glucose (FBG), oral glucose tolerance test (OGTT), and glycated hemoglobin (HbA1c). However, these methods are not without limitations. Measurements of FBG and OGTT can be influenced by transient physiological conditions such as stress, illness, or short-term dietary changes, potentially affecting their accuracy.²⁷ Meanwhile, HbA1c may yield less reliable results in certain clinical settings, including anemia, renal disorders, and hemoglobinopathies, which can lead to misclassification of glycemic status. These limitations highlight the need for more reliable biomarkers to assess early disturbances in glucose metabolism.¹⁷

One biomarker that has gained attention in recent years is 1,5-Anhydroglucitol (1,5-AG). This marker reflects short-term fluctuations in glycemic levels and has been reported to be more responsive than HbA1c in detecting postprandial hyperglycemia. Due to its relative stability and ease of measurement, 1,5-AG has been proposed as a potential indicator of early glycemic abnormalities, including prediabetes.^{8–10} Previous studies have shown that circulating levels

of 1,5-AG decrease as glucose excursions increase, supporting its role as a complementary marker of early glycemic dysregulation, although its routine clinical application remains limited.

Based on these considerations, this study aimed to assess the diagnostic accuracy of plasma 1,5-Anhydroglucitol (1,5-AG) in identifying prediabetes among obese young adults. By comparing its performance with HbA1c as the reference standard, this study seeks to further clarify the potential utility of 1,5-AG as an additional biomarker for early detection in high-risk populations.

13 METHODS

Study Design and Setting

This diagnostic accuracy study was conducted over a six-month period at Dr. M. Djamil General Hospital, Padang, Indonesia. The study aimed to evaluate the diagnostic performance of plasma 1,5-Anhydroglucitol (1,5-AG) for detecting prediabetes among obese young adults.

Participants

The study included obese adults aged 18–40 years who visited the Internal Medicine outpatient clinic throughout the study period. Participants were recruited using incidental sampling.

Participants were eligible if they were between 18 and 40 years of age, had a body mass index (BMI) above 25 kg/m², and provided consent to take part in the study. The age range of 18–40 years was selected to represent young adults, a population increasingly affected by obesity-related metabolic disorders and at risk of early-onset dysglycemia. Exclusion criteria included a prior history of prediabetes or diabetes mellitus, kidney disorders, the use of sodium-glucose cotransporter-2 (SGLT2) inhibitors or corticosteroids, and erythrocyte abnormalities. Baseline characteristics were recorded, comprising demographic and clinical variables such as age, sex, family history of diabetes, waist circumference, body weight, height, and body mass index.

Data Collection and Laboratory Examination

Venous blood samples (6 mL) were obtained from each participant for the measurement of glycated hemoglobin (HbA1c) and plasma 1,5-Anhydroglucitol (1,5-AG) levels. These measurements were used to assess the glycemic status of the participants.

All laboratory examinations were performed at the Clinical Pathology Laboratory of Dr. M. Djamil General Hospital, Padang. HbA1c was measured using a standardized NGSP-certified laboratory method, while plasma 1,5-Anhydroglucitol levels were measured using a commercially available enzymatic assay kit, following the manufacturer's instructions. All samples were analyzed under standardized laboratory conditions, with acceptable intra- and inter-assay variability according to laboratory quality control standards.

Reference Standard and Diagnostic Analysis

HbA1c was used as the reference standard for the classification of prediabetes, due to its widespread clinical acceptance, despite recognized limitations in certain clinical conditions. Descriptive analysis was performed to summarize the baseline characteristics of the participants.

The diagnostic accuracy of plasma 1,5-Anhydroglucitol was assessed by determining sensitivity, specificity, and both positive and negative predictive values (PPV and NPV) based on a 2 × 2 contingency table comparing 1,5-AG measurements with HbA1c-based classification.

Ethical Considerations

This study was approved by an institutional health research ethics committee (Approval No. DP.04.03/D.XVI.XI/622/2023). All participants received detailed information regarding the study and provided written informed consent prior to participation. Participation was voluntary, and all study-related costs were covered by the researchers.

RESULTS AND DISCUSSION

Baseline Characteristics of the Study Population

A total of 143 obese young adults were included in this study. The participants' baseline demographic and clinical profiles are summarized in Table 1. Overall, the study population consisted predominantly of young adults with obesity, most of whom demonstrated central adiposity based on waist circumference measurements.

Table 1. Baseline Characteristics of the Study Population

Characteristics	n (%)	Mean (SD)	p
Gender			0.656
Male	79 (55.2)		
Female	64 (44.8)		
Age (years)		30.4 (3.88)	0.007
18–24	12 (8.39)		
25–29	29 (20.28)		
30–34	86 (60.14)		
35–40	16 (11.19)		
Occupation			0.021
Resident Doctor (PPDS)	129 (90.21)		
General Practitioner	14 (9.79)		

Body Mass Index (kg/m²)		29.97 (3.49)	0.949
Obesity I (25–29.9)	83 (58.05)		
Obesity II (30–34.9)	51 (35.66)		
Obesity III (≥35)	9 (6.29)		
Family History of Diabetes			0.033
Yes	44 (30.8)		
No	99 (69.2)		
Waist Circumference			0.615
Male		94.47 (10.42)	
>90 cm	73 (92.41)		
≤90 cm	6 (7.59)		
Female		99.97 (9.55)	
>80 cm	62 (96.88)		
≤80 cm	2 (3.12)		
Hemoglobin A1c (%)		5.08 (0.66)	
1,5-Anhydroglucitol (µg/mL)		7.89 (4.76)	

Distribution of Prediabetes Based on HbA1c

HbA1c levels were measured in all participants. The frequency distribution of prediabetes based on HbA1c classification is shown in Table 2.

Table 2. Frequency Distribution of Prediabetes Based on HbA1c

HbA1c Category	n	%
Non-Prediabetes	123	86
Prediabetes	20	14
Total	143	100

Among the 143 participants, 20 individuals (14.0%) were classified as having prediabetes.

Distribution of Prediabetes Based on Plasma 1,5-Anhydroglucitol

The frequency distribution based on plasma 1,5-Anhydroglucitol levels is presented in Table 3.

Table 3. Frequency Distribution of Prediabetes Based on Plasma 1,5-Anhydroglucitol

1,5-AG Category	n	%
Non-Prediabetes	94	65.7
Prediabetes	49	34.3
Total	143	100

Diagnostic Performance of Plasma 1,5-Anhydroglucitol

The diagnostic performance of plasma 1,5-Anhydroglucitol for detecting prediabetes was evaluated using HbA1c as the reference standard. The cross-tabulation results are presented in Table 4.

Among the 143 participants, 18 cases were identified as true positives, 31 as false positives, 2 as false negatives, and 92 as true negatives.

Table 4. Diagnostic Performance of Plasma 1,5-Anhydroglucitol

	HbA1c Prediabetes	HbA1c Non-Prediabetes	Total
1,5-AG Prediabetes	18 (TP)	31 (FP)	49
1,5-AG Non-Prediabetes	2 (FN)	92 (TN)	94
Total	20	123	143

The findings of this study indicate that plasma 1,5-Anhydroglucitol demonstrates strong diagnostic potential for identifying prediabetes, particularly due to its high sensitivity and excellent negative predictive value. These results suggest that this biomarker is especially useful in ruling out prediabetes among obese young adults, as individuals with normal 1,5-AG levels are unlikely to have impaired glycemic status. However, the relatively low positive predictive value observed indicates that a considerable proportion of individuals identified as positive may not truly have prediabetes, thereby limiting its use as a standalone diagnostic tool.

From a clinical standpoint, the implementation of plasma 1,5-AG testing may still face certain challenges. The availability of this assay is not yet universal across healthcare facilities, and the associated costs are generally higher compared to commonly used markers such as HbA1c. These factors may restrict its application in routine clinical practice. Therefore, despite its promising performance, 1,5-AG is currently more appropriately positioned as an adjunctive biomarker rather than a primary diagnostic method.

In this study, the proportion of male participants slightly exceeded that of females. This pattern is consistent with several previous reports that have observed a higher prevalence of obesity among young adult males. Nevertheless, other studies have demonstrated comparable prevalence rates between sexes, suggesting that differences in obesity patterns may be influenced by demographic

characteristics, lifestyle behaviors, and environmental factors.^{11–13}

³⁷ The mean age of participants in this study was 30.4 years, indicating that metabolic disturbances related to obesity may emerge relatively early in adulthood. This observation aligns with earlier findings highlighting the increasing occurrence of prediabetes among younger populations. Reduced levels of physical activity and changes in dietary habits during adolescence and early adulthood are often implicated as contributing factors to this trend.^{11, 16, 17}

²⁴ Obesity plays a critical role in the development of insulin resistance and impaired glucose metabolism. Differences in fat distribution between males and females may contribute to variations in metabolic risk. Visceral fat accumulation, which is more common in males, is strongly associated with insulin resistance and adverse metabolic outcomes, whereas peripheral fat distribution, more frequently observed in females, tends to be linked with a relatively lower cardiometabolic risk.^{14, 15}

The average body mass index of participants in this study indicated that most individuals fell within the category of class I obesity. Although BMI is widely used as a measure of obesity, it ³⁴ does not distinguish between fat mass and lean body mass. As a result, waist circumference is often considered a more informative indicator of central adiposity and cardiometabolic risk. In this study, the mean waist circumference values suggested that the majority of participants exhibited ²⁷ central obesity, which is known to be ³³ associated with an increased likelihood of developing metabolic disorders, including prediabetes and type 2 diabetes.^{20, 21}

⁴⁶ A family history of diabetes was present in nearly one-third of participants and appeared to be associated with the occurrence of prediabetes. Previous evidence suggests that individuals with such a background have a higher lifetime risk of developing glucose metabolism disorders, likely due to the combined influence of genetic susceptibility and shared environmental exposures.^{22–25}

The prevalence of prediabetes in this study, based on HbA1c classification, was 14%, which is lower than that reported in several other populations. This difference may be attributed to variations ³⁶ in study design, population characteristics, lifestyle factors, and diagnostic criteria applied across studies.^{26, 27}

The physiological mechanism underlying the relationship between 1,5-AG and glycemic control is closely related to renal glucose handling. Under normal conditions, ²¹ 1,5-AG is filtered by the kidneys and almost completely reabsorbed in the renal tubules. During episodes of hyperglycemia, elevated glucose concentrations compete with 1,5-AG for reabsorption, ¹⁸ resulting in increased urinary excretion and reduced plasma levels of 1,5-AG. This mechanism explains the inverse ⁴⁴ relationship between plasma 1,5-AG levels and glucose excursions observed in individuals with impaired glucose regulation.³⁰

The high sensitivity and negative predictive value demonstrated in this study reinforce the potential role of plasma 1,5-AG in early detection strategies. Although the specificity observed was lower than that reported in some previous studies, the ability of this biomarker to identify individuals without prediabetes remains clinically valuable. Previous studies have also linked lower 1,5-AG levels with

reduced insulin sensitivity and pancreatic β -cell dysfunction, further supporting its relevance in early metabolic assessment.

² Despite these findings, several limitations should be acknowledged. The study was conducted in a single center with a relatively limited sample size, which may affect the generalizability of the results. In addition, body composition was not evaluated using more advanced techniques such as bioimpedance analysis or dual-energy X-ray absorptiometry, which could have provided a more detailed assessment of adiposity. Future studies involving larger and more diverse populations, as well as more comprehensive metabolic evaluations, are required to further validate the diagnostic utility of plasma 1,5-Anhydroglucitol in prediabetes detection.

CONCLUSIONS AND SUGGESTIONS

The findings indicate that plasma 1,5-AG demonstrates high sensitivity and excellent negative predictive performance in detecting prediabetes among obese young adults. Using HbA1c as the reference standard, plasma 1,5-AG showed a sensitivity of 90.0%, specificity of 74.8%, along with positive and negative predictive values of 36.7% and 97.9%, respectively. Overall, these results indicate that plasma 1,5-AG ³⁰ may serve as a complementary biomarker for identifying individuals at increased risk of prediabetes within this population. However, due to its moderate specificity and low positive predictive value, plasma 1,5-AG should be interpreted alongside established glycemic markers. ²⁹ Further studies with larger populations are needed to confirm the diagnostic role of plasma 1,5-Anhydroglucitol in prediabetes detection.

⁵ ACKNOWLEDGEMENT

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